



Impact assessment of climate-smart livestock practices adoption on farmers' food security in Benin

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Abstract

This study aims to 1) analyse the factors influencing the adoption and the intensity of adoption of climate-smart livestock practices among farmers in Benin and 2) to assess the impact of climate-smart livestock practices on food security. For this, data from 460 farmers selected randomly were collected face-to-face using the application Kobocollect on smartphones. Firstly, a Logit model was chosen to analyse factors influencing the adoption of climate-smart livestock practices. Secondly, we chose Generalized Poisson Regression (GPR) instead of Standard Poisson Regression using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) tests to analyse factors influencing the intensity of CSL practices adoption. Finally, the propensity score matching method, combined with inverse probability weight and regression adjustment, was used to analyze the impact of CSL on food security. The results showed that factors such as experience in livestock, the cattle herd size, and belonging to a cooperative influence positively and significantly the adoption of CSL practices at 5 %. Also, education (SDG4) and agriculture advices are factors influencing CSL adoption positively and significantly at 1 %. For the intensity of CSL adoption, factors like the contact with extension services, agriculture advices, and agriculture as the first activity (SDG2) influence this intensity. On the other hand, the results show that climate-smart livestock adoption improves the household food insecurity access score (HFIES) but doesn't have any influence on household dietary diversity score and vitamin A consumption. Policymakers to improve food security can encourage farmers to widely adopt CSL

The policymakers can be based on cooperative existing and extension services to promote the climate-smart livestock practices, thereby enhancing farmers' resilience to climate change.

Keywords: Adoption, Intensity of adoption, Climate-smart livestock practices (CSL), Poisson Regression, and Generalized Poisson Regression (GPR), Food security, Propensity score matching (PSM).

1. Introduction

Agriculture in sub-Saharan Africa is mostly rain-fed and managed by smallholder farmers (Andoh & Kaba, 2025). Agriculture and food security in this region of Africa, already vulnerable due to socio-economic challenges and its reliance on rain-fed agriculture, are facing numerous challenges posed by climate change. Rising temperatures alter precipitation patterns, droughts, and floods, which impact agricultural productivity, exacerbating food insecurity over time (Missanga & Mmbando, 2025; Omokpariola & al., 2025). There is therefore an urgent need to adapt crop and livestock systems to make them more resilient to climate trends and extreme events. At the same time, it is necessary to reduce emissions from the agricultural sector and increase carbon sequestration to mitigate longer-term climate impacts (Debaeke et al., 2025). The global effect of climate variability and change on agricultural production systems is a pressing concern with far-reaching implications (Ningi et al., 2025). This phenomenon of climate change has a direct impact on the current and future agricultural productivity (Li et al., 2025). Climate change has also direct implications on agriculture globally and especially in sub-Saharan Africa, where many small-scale farmers are dependent on this activity for their livelihoods (Akram et al., 2025).

Several studies have addressed the consequences of climate change on agriculture and livestock (Cheng et al., 2022; Kone et al., 2024; Mahdavian et al., 2025). Climate change poses a significant threat to livestock production in Sub-Saharan Africa, with major impacts on food security, poverty, rural livelihoods, and greenhouse gas emissions (Duku et al., 2025; Ngongolo & Gayo, 2025). The livelihoods of smallholders, such as pastoralists and agro-pastoralists, in Sub-Saharan Africa (SSA), are threatened due to their exposure and sensitivity to the surrounding environment, including increases in temperature and changes to rainfall patterns resulting from the effects of climate change (Stagner, 2025). The livestock farmers' response to the climate change phenomenon is very low due to a lack of scientific and context-based evidence (Bontsa et al., 2024).

Climate-smart agriculture (CSA) is widely acknowledged as a way of promoting sustainable agriculture and food security in Sub-Saharan Africa (Yiridomoh et al., 2025). Climate-smart agriculture (CSA) has the potential to mitigate the adverse consequences of climate change and directly influence household well-being. However, CSA concepts are still not widely applied at the farm level in poor nations (Daudu et al., 2025). Livestock-focused climate-smart (CS) technologies, including grazing and manure management, methane-reducing feed additives, selective breeding, and genetic engineering, aim to reduce emissions, increase productivity, and improve resilience to climate change (Khanal & Dhoubhadel, 2025a).

Climate-smart agriculture (CSA) has been widely leveraged as an approach to achieving both food security and climate change outcomes through suites of interventions that maximize synergies and reduce tradeoffs among three pillars: productivity, climate change resilience, and climate change mitigation (Germer et al., 2023). Livestock, being an interwoven part of the ecological balance, is also witnessing measurable effects of climate change. Reduction in feed resource efficiency, outbreak of diseases, heat stress, breeding problems, reduction in production attributes, etc., are the direct measurable effects of climate change on livestock (Singh et al., 2024). Livestock-focused climate-smart technologies aim to reduce emissions, increase productivity, and improve resilience to climate change (Khanal & Dhoubhadel, 2025b).

Climate-smart agricultural practices are necessary across West Africa and sub-Saharan Africa. Climate-Smart Agriculture (CSA) is an approach that provides a conceptual basis for assessing the effectiveness of agricultural practice change to support food security under climate change (Amole & Ayantunde, 2016). Climate-smart agriculture (CSA) is an approach that identifies actions needed to transform and reorganize agricultural systems to effectively support agricultural development and ensure food security in the face of climate change (Moutouama et al., 2022). Climate-smart agriculture (CSA) is an appropriate adaptation approach to climate change. It allows producers to improve production and enhance food security despite the impacts of climate change, while limiting agricultural-related causes (Midjangninou et al., 2024).

In Benin, as in most developing countries, growers are increasingly faced with the need to adapt their cropping systems to changing climatic circumstances (Yegbemey et al., 2014). Climatic events and their effects are perceived by local populations and adaptations are developed to cope with them (Kate et al., 2015). Climate change is increasing in Benin and affecting the rural environment and agrarian balance, causing a decline in productivity and efficiency (Sodjinou and Hounkponou, 2019), cattle farming occupies a significant proportion of the population and is one of the major economic activities, particularly in northern Benin, where there are different groups of farming systems in the communes of Gogounou and Kalalé (Gounou and Yabi, 2020; Zakari et al., 2015; Dia, 2014).

Food insecurity can worsen diet quality and consequently increase the risk of various forms of malnutrition, potentially leading to undernutrition as well as overweight and obesity, which can have negative health consequences (Morales & Berkowitz, 2016). Low-income countries rely more on staple foods and less on fruits and vegetables and animal source foods than high-income countries. Only in Asia, and globally in upper-middle-income countries, are there enough fruits and vegetables available for human consumption to be able to meet the FAO/WHO recommendation of consuming a minimum of 400 g/person/day (IFAD, 2017).

This study is organised as follows: the subsequent section presents the theoretical framework employed in this study. The following sections are dedicated to methodology, results, and discussion.

2. Methodology

The study focuses on Benin, a West African nation spanning 114,763 km² between longitudes 1° and 3°40' E and latitudes 6°30' and 12°30' N. It is bounded by Togo to the West, Burkina Faso and Niger to the North, Nigeria to the East, and the Bight of Benin to the south. Benin is home to three major cities: Cotonou, the economic capital and the most populous urban center; Porto-Novo, the official administrative capital, known for its rich cultural heritage; and Parakou, the principal city in the northern region and a vital commercial hub. Our study was conducted in three communes: Gogounou, Malanville, and Karimama (Figure 1). These communes were selected due to the high importance of livestock, agro-ecological diversity, and the effects of climate change that are mostly endured.

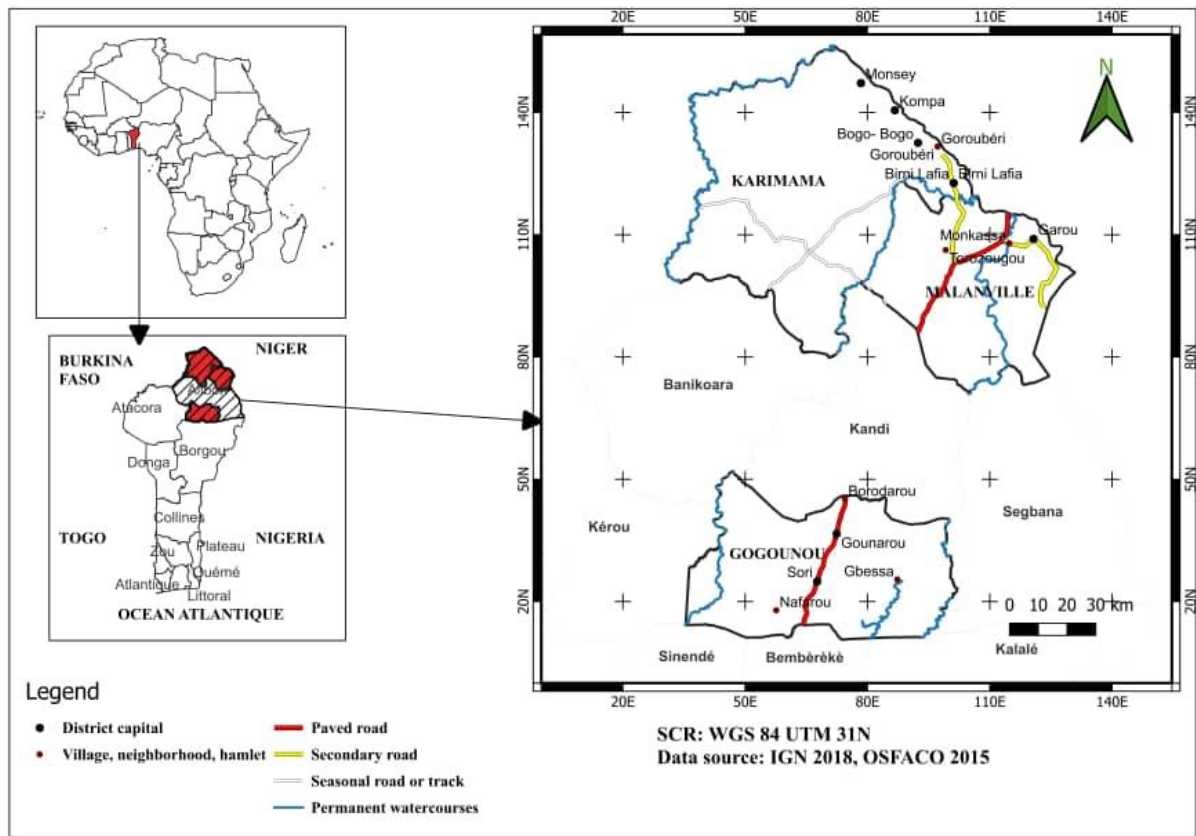


Figure 1: Map of study area

3. Data collection

This study utilized cross-sectional data from a household survey conducted in the three communes of Malanville, Gogounou, and Karimama. Two groups of farmers were selected in this research. The first group is farmers who adopted climate-smart livestock practices, and the second group is farmers who didn't adopt. A total of 45 villages engaged in climate-smart livestock in these three communes to obtain diversity data for the study. The climate-smart livestock practices are: introduction of improved breeds, crossbreeding with local breeds for good disease resistance, building up food reserves for the dry season (hay, silage, etc.), using new food sources (shrubby legumes, rice straw, etc.), growing resistant forage varieties, practicing seasonal livestock mobility, and medical prophylaxis. Farmers who use at least one of these climate-smart livestock practices are considered adopters.

These data were obtained from 460 livestock households through face-to-face interviews using the Kobocollect software between July and August 2025. Data from 89 farmers who practice climate-smart livestock and 370 who don't implement at least one of the climate-smart livestock practices. The gathered data include information on, e.g., household characteristics, climate-smart livestock practices, household assets, livestock keeping, and food & nutritional security parameters. The household head and/or his/her spouse were asked questions about their food security and different food groups consumed in the household.

The data was collected using the KoboToolbox platform via the KoboCollect application on a smartphone. The interviews were conducted mostly in the local language.

4. Theoretical Framework model specification

In this study, random utility maximization theory was used to assess factors influencing the adoption and intensity of climate-smart livestock practices. This theory stipulates that farmers adopt technologies, practices, or innovations that maximize their expected utility under conditions of uncertainty (Hategekimana et al., 2025; Ng'ombe et al., 2025). So, the decision of farmers to adopt climate-smart livestock is influenced by socioeconomics, institutional disposition, and farming constraints. The farmer's utility maximization is assessed at two levels: first, in the binary decision to adopt any climate-smart livestock practices, and second, in a count of climate-smart practices adopted.

To test this theory empirically in our study, two steps of an analytical approach are used: the binary model (logit model) and the Poisson regression, respectively, to analyse the factors that influence adoption and the intensity of adoption.

4.1. Logit model

The Logit model analysis was used to examine the factors influencing the adoption of CSL technologies in beef breeding. This involved analyzing the relationship between a binary dependent variable and a set of independent variables, including both binary and continuous variables (Sanogo et al., 2023).

$$y_i = \begin{cases} 1 & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases} \quad (\text{eq } 1)$$

4.2. Poisson Regression

The first regression used in this research is the Poisson regression. It aims to identify the factors that influence the number of practices adopted by farmers.

The factors affecting the adoption of Climate-smart livestock degree practices will be analyzed empirically using Poisson regression. Thus, the Poisson regression analysis is appropriate for analyzing such data, counted data. The Poisson regression model can be estimated using the quasi-maximum likelihood approach because it overcomes the problems associated with over-dispersion, which leads to grossly deflated standard errors and inflated t-statistics in the output, and is consistent under this condition (Abate & Addis, 2021).

$$Pr(Y = y) = \frac{e^{(-\mu)} \mu^y}{y!}, y = 0, 1, 2, \dots, n$$

5. Estimation of Food Security

Food security is defined as the situation, “when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life” (Food Summit, 2009). In this paper, food security is apprehended by a combination of two measures: the Household Food Insecurity Access Scale (HFIES), the household dietary diversity (HDDS), and the Vitamin A consumption. The last two metrics were measured during the last seven days, and the day before the interview.

According to Aïhounton & Henningsen, (2024), the household head and/or his/her spouse were asked whether their household had experienced the following situations during the last 12 months because of a lack of money or other resources:

- You were worried your household would not have enough food to eat.
- Your household was unable to eat healthy and nutritious food.
- Your household ate only a few different kinds of food.
- Your household had to skip a meal.
- Your household ate less than you thought it should.

Your household ran out of food.

Your household was hungry but did not eat.

Your household went without eating for a whole day

HFIES is assessed by four ordered answers: 0 = Never, 1 = Rarely, 2 = Sometimes, and 3 = Often. This allows us to obtain the original HFIES, which range from 0 to 8 (i.e., the number of answers that are not 0), and a more detailed HFIES that ranges from 0 to 24 by summing up the answers to the eight questions.

Also, in this paper, the household dietary diversity score suggested by FAO (2011) as an indicator for quality aspects of food security.

Cereals (bread, maize, sorghum, oats, etc.)

White tubers and roots (white potatoes, white yams, white cassava, or other roots)

Vitamin A-rich vegetables and tubers (carrots, sweet potatoes, sweet red peppers)

Dark green leafy vegetables

Other vegetables

Vitamin A-rich fruits (mango, papaya, dried peach), and 100% fruit juice made from these

Other fruits

Organ meat (liver, kidney, heart or other organs or blood-based foods)

Flesh meats

Eggs

Fish and seafood

Legumes, nuts, and seeds (seed: all other seeds except cereals)

Milk and milk products

Oils and fats

Sweets

Spices, condiments, beverages (beverages: coffee, tea, alcoholic beverages).

6. Propensity score matching

There are two methods of impact evaluation used to assess the outcome of an intervention or program: experimental and quasi-experimental methods. Experimental methods are widely criticized in the literature. These criticisms mainly concern: the effect of the experiment, the nature of the program offered, the composition of the participants, and the behavior of non-participants (Heckman, 2010). The quasi-experimental or non-experimental method is an alternative to the experimental method that seeks to create a control group defined by a counterfactual (Wonde et al., 2022). The group inequality can now be adjusted and rectified using a balancing score to fix the imbalance between the groups (threatened and unthreatened) using PSM, where confounding factors can be identified. The balance score indicates that the treatment group, which is the adoption of CSL practices, should not affect the observed variable (Fente et al., 2025). The average treatment effect for the treated (ATET) represents the difference in outcomes between farmers who have adopted CSL practices and those who have not (Houdji & Affossogbe, 2026).

In addition to propensity score matching, we used the Regression adjustment. The standard regression adjustment for the propensity score is an alternative to the aforementioned propensity score methods (Vansteelandt & Daniel, 2014). This doubly robust property makes IPWRA particularly suitable for robustness checks and provides additional confidence in the stability and reliability of the estimated impacts (Ouédraogo et al., 2026).

7. Results and Discussion

7.1. Sociodemographic characteristics of the sample

Table 1: Sociodemographic Characteristics

Variables	Mean	Standard Deviation	Minimum	Maximum
Age	39.532	9.761	13	77
HH Size	6.678	3.530	1	25
Secondary Activity	0.596	0.491	0	1
Sex	0.922	0.269	0	1
Livestock Experience	7.843	7.209	1	56
Cattle Head size	9.229	15.371	1	130
Extension	0.192	0.394	0	1
Belong to cooperative	0.501	0.501	0	1
length of cooperative belongs	4.900	3.919	2	45
Knowledge of CSA	0.427	0.495	0	1
Practice at least one of CSLs	0.194	0.396	0	1

Table 1 shows that, on average, the smallholders were 39,5 years with approximately 10 years as variation around the mean. The older smallholder was 77 years old, and the youngest was 13 years. In the sample, we have an average of 7 persons per household with a variation of 3 persons around this mean. The livestock mean experience in this area study is 7 years, and the oldest livestock practitioner cumulate 56 years in the livestock. In the sample, the farmers have on average 9 cattle herds. The results show that 92 % of farmers are men. We find that the majority (52%) of farmers have a secondary activity. The findings also show that 50 % of farmer taken account in this study belong to a cooperative since in mean 5 years. We also find that 43 % of farmers have CSA knowledge, and only 19 % practice at least one of the CSAs.

7.2. Determinant of Climate-smart livestock practices adoption

Logit Model

Table 4: Parameter estimates of Logit model for the factors determining the adoption of climate-smart livestock practices

SDG 13 Adoption	Coef.	Std. Err.	Margin effects	Std. Err
Household size	-0.190***	0.065	-0.019***	0.006
Secondary activity	-0.924**	0.399	-0.090**	0.038
Experience in livestock	0.053**	0.026	0.005**	0.003

SDG 13 Adoption	Coef.	Std. Err.	Margin effects	Std. Err
Cattle herd size	0.022**	0.010	0.002**	0.001
Access to a loan	1.126**	0.570	0.110**	0.055
Extension services	0.161	0.468	0.016**	0.046
Belongs to a cooperative	0.884**	0.379	0.086**	0.036
Education (SDG4)	0.111***	0.039	0.011***	0.004
Agriculture advices	2.786***	0.355	0.272***	0.025
Sex (SDG5)	0.482	0.606	0.047*	0.059
Agriculture as 1st activity	0.993	0.839	0.097*	0.081
Livestock as a secondary activity	1.924*	0.999	0.188**	0.096
Constant	-4.645***	1.114	-0.019***	0.006
Number of observations	460		Prob > chi2	0.00
LRchi2(11)	141.40		Pseudo R2	0.3132

The model shows that 31.32 % of the variables introduced in the model explain the adoption of at least one climate-smart agriculture practice. The model is globally significant at 1 %. The variables, such as household size, the possession of secondary activity, the number of years of experience in livestock, the cattle herd size, access to loan, cooperative membership, SDG4, and livestock as principal activity, have a significant influence on agriculture-smart practices adoption. But the variables such as experience, contact with extension services, sex, and agricultural as the first activity haven't influenced livestock-smart practices.

The model logit shows that household size significantly and negatively affects the adoption of climate-smart livestock practices at 1 %. The increasing of one person and the household will reduce the climate-smart livestock adoption probability to 1.9 %.

Also, the holding of a secondary activity negatively at 5 % affects the adoption probability of climate-smart livestock. So, the possession of secondary activity decreases the climate-smart livestock adoption probability of 9 %. At the same time, livestock as the secondary activity influences positively and significantly at 5 % the adoption probability of Climate-smart livestock to 18.8 %.

On the other hand, the experience of in livestock has a positive and significant at 5 % influence on climate-smart livestock. The growth of 1 year of experience in livestock increases the adoption probability of 0.5 %.

Another variable in the model that influences the probability of climate-smart adoption is cattle herd size. So, the increase of 1 cattle herd grows this probability to 0.2 %. After access to loan enhance significantly at 5 %, the climate-smart adoption probability increased to 11 %. The belonging to a cooperative also significantly ameliorates the climate-smart livestock probability adoption at 5% of significantly to 8.6 %.

SDG4, that's represented the level of education, influences the adoption of climate-smart livestock at 1 %. Indeed, the growth of one successful education year increases the adoption of CSL of 1.1 %.

The access to agriculture advices ameliorates significantly at 1 % the adoption of CSL to 27.2 %. Agricultural advice increases the probability of CSL adoption. So, when the smallholder has agricultural advice, this probability increases positively and significantly to 27.2 % at 1 %.

7.3. Determinants of Climate-smart livestock practices adoption intensity

To identify the factors that influence the intensity adoption of CSL, both the generalized Poisson model and the standard Poisson model were used. Then, diagnostic tests were conducted to determine the most appropriate model for the analysis. Based on model selection criteria, taking into account the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), the generalized Poisson model was favored in this case. So, the results discussion will be relative to Generalized Poisson Regression. The findings of the generalized Poisson regression model show many variables that influence the intensity of climate-smart livestock.

The cattle herd size significantly and negatively at 10 % the adoption intensity of climate-smart livestock. This result is in contradiction with that of Mujeyi et al., (2021), who find that farmers who adopted Climate-smart agriculture practices have higher herd sizes. The results show that the increase of one unit of cattle size will decrease the number of climate-smart livestock practices adopted. This adoption probability will decrease to 87 %.

Access to loans has significantly and positively influenced the intensity of CSL adoption at 5 %. When the farmer contracted a loan, he was mostly open to adopting more CSL practices. This result concurs with other researchers who find that access to credit has a positive and significant influence on the climate-smart agriculture level adoption (Ojo et al., 2023).

The contact with extension services has a positive and significant impact on the intensity of CSL practice adoption. So, the contact extension services will facilitate the CSL practices adoption at 5 %. Because the extension will sensitize farmers on the necessity to adopt CSL practices for climate change effects mitigation and its benefits. Makate et al., (2019) found that access to either credit or extension services has an impact on climate-smart agriculture progress adoption. This result confirms the findings of this paper that contact with extension services has a significant and positive effect on CSL practices adoption.

In the same vein, agricultural advice as extension services influences the number of CSL practices adopted. This influence is positive and significant at 1 %. Other findings obtained the same result that agricultural advice influences CSA adoption (Gemtoui et al., 2024).

In fine, the SDG2 that represents agriculture as the first activity has a positive and significant impact on the number of CSL practices adoption at 1 %. The possession of SDG2 as the first activity increases the probability of CSL adoption to 0.03 %.

The government, through its services as a territorial agency of agricultural development (ATDA), and the agricultural departmental direction authorised projects/programs and NGO to promote the climate-smart livestock practices. Also, in all regions in the country, the government has an extension service agent to disseminate its instruments and policies. Moreover, the cooperatives are the best way to reach a lot of farmers to promote a new policy.

The gaps identified in institutional coordination are: lack of clear objectives, defined strategy, communication, and information on climate-smart agriculture or livestock practices, poor allocation of tasks and responsibilities, and politicised subsidies.

	Standard Poisson Regression		Generalized Poisson Regression	
Intensity SDG 13 Adoption	Coefficient	Margins effect	Coefficient	Margins effect
Household size	-0.057** (0.026)	-0.945** (0.006)	-0.257 (0.217)	-0.773 (0.168)
Secondary activity	1.294*** (0.326)	3.649*** (0.038)	2.012 (1.371)	7.478 (10.251)

	Standard Poisson Regression		Generalized Poisson Regression	
Experience in livestock	0.001 (0.017)	1.001 (0.003)	0.026 (0.083)	1.026 (0.085)
Cattle herd size	-0.004 (0.008)	-0.996 (0.001)	-0.139* (0.076)	-0.870* (0.066)
Access to a loan	0.548*** (0.21)	1.729*** (0.055)	5.658* (3.122)	286.515* (894.437)
Extension services	0.747*** (0.23)	2.110*** (0.046)	5.651** (2.63)	284.500** (748.212)
Belongs to a cooperative	0.528* (0.277)	1.695* (0.036)	1.577 (1.153)	4.841 (5.581)
Education (SDG4)	0.029 (0.024)	1.029 (0.004)	-0.128 (0.173)	-0.880 (0.152)
Agriculture advices	1.781*** (0.263)	5.934*** (0.025)	4.882*** (1.699)	131.931*** (224.086)
Sex (SDG5)	-0.692** (0.273)	-0.501 (0.059)	1.249 (1.636)	3.488 (5.705)
Agriculture as 1st activity	-0.878*** (0.244)	-0.416*** (0.081)	-7.973* (3.062)	-0.0003*** (0.001)
Livestock as a secondary activity	0.362 (0.338)	1.436 (0.096)	-5.412 (3.358)	0.004 (0.015)
Constante	-2.502 (0.489)		0.815 (2.542)	
Number of observations	460		Prob > chi2	0.00
LRchi2(11)	284.33		Pseudo R2	0.3284
AIC	569.0168		378.4564	
BIC	622.2332		435.7664	

Note: ***, **, and * indicate 1 %, 5 %, and 10 % significance levels, respectively.

7.4. Syntheses SDG synergies and trade-offs

	SDG 13: Climate-smart livestock practices
SDG 2 (Zero hunger)	+
SDG3 (Good Health and Well-being)	+
SDG4 (Quality Education)	0
SDG5 (Gender Equality)	+
SDG 15 (Life and land)	+

Table: SDG synergies

The table shows the interactions between different SDGs and SDG 13, which is climate-smart livestock in this research. So, SDG 13 has a positive effect on SDG 2, SDG 3, SDG 5, and SDG 15. The results of this study reveal firstly that household who adopted CSL increase their food security. We see a positive effect between these indicators (SDG 2 and SDG 13). In the same way, the adoption of climate-smart livestock ameliorates the household well-being through their Dietary Diversity Score. As in the first case, there is a positive interaction between SDG 13 and SDG 3. Thirstily, the gender equality (SDG 5) has a positive influence on SDG 13 because the model shows that when the household head is a man, that have an negative effect on adoption. So, when the head is a woman, that influences the CSLs' practices adoption positively. In fine, the adoption of SDG 13 ameliorates life and land because of promotion practices that don't impact fauna and flora.

7.5. Treatment-effects estimation (PSM) of CSL on food insecurity

Table 1: Impact of climate-smart livestock on HFIES

HFIES	Coefficient	Standard error	z
ATET			
Adopt at least one CSL practice (Yes vs No)			
Nearest-neighbour matching	-2.131***	0.769	-2.77
Regression adjustment	-1.483***	0.558	-2.66
Inverse-probability weights	-1.821***	0.582	-3.130

Number of observations = 460

The results of PSM with the nearest-neighbour matching method, combined with two other methods, such as regression adjustment and inverse-probability weights, show that the adoption of climate-smart livestock practices has a positive and significant effect on food security. So, when the farmer adopted the CSL practices, this decreased by 2.131 the HFIES with the nearest-neighbour matching method, by 1.483 the score of food insecurity for the RA method, and by 1.821 the score of household food insecurity for inverse-probability weight. In conclusion, the farmers adopters of CSL practices are more secure in food security than those who do not adopt. Adoption of improved and resilient livestock breeds reduces household food insufficiency (Radeny et al., 2022a). In the same way, Mujeyi et al., (2021) found that climate-smart agriculture improves the food security score by 1.492.

7.6. Treatment-effects estimation (PSM) of CSL on dietary diversity during one week

Table 2: Climate-smart livestock impact on HDDS during 7 days

HDDS_7Days	Coef,	Standard error	z
ATET			
Adopt at least one CSL practice (Yes vs No)			
Nearest-neighbor matching	0.11	0.34	0.35
Regression adjustment	0.192	0.280	0.68
Inverse-probability weights	0.373	0.296	1.260

Number of observations = 460

Regarding the household dietary diversity during the last seven days, results show that for three methods, such as nearest-neighbor matching, regression adjustment, and inverse-probability weights, the adoption of CSL did not significantly impact the score of household dietary diversity during the last seven days. This result is the opposite of that found by Radeny et al. (2022b). For them, the adoption of improved and resilient livestock breeds (including Red Maasai sheep and Galla goats) improved household dietary diversity by 38%. Climate-smart agriculture practices adoption significantly enhances Household Dietary Diversity Scores (HDDS) by 30% (Omotayo et al., 2025).

7.7. Treatment-effects estimation (PSM) of CSL on household dietary diversity score during 24H

Table 3: Climate-smart livestock impact on HDDS during 24 hours

HDDS_24H	Coef,	Standard error	z
ATET			
Adopt at least one CSL practice (Yes vs No)			
Nearest-neighbor matching	0.63	0.36	0.17
Regression adjustment	-0.050	0.271	-0.190
Inverse-probability weights	0.011	0.293	0.040

Number of observations = 460

The impact of climate-smart livestock practices adoption on household dietary diversity score during the last 24 hours is not significant, as presented in the results. So, the adoption of climate-smart livestock practices does not affect a household's economic ability to access a variety of foods. This finding is contrary to that of Debie, (2025), who concluded that climate-smart agriculture enhances the household dietary diversity scale. The Household Dietary Diversity Score analysis reveals that Climate-smart agriculture practice adopters achieve markedly higher dietary diversity scores (Mnukwa et al., 2026). This finding is also different from ours, while we focus on climate-smart livestock practices instead of climate-smart agriculture practices.

7.8. Treatment-effects estimation (PSM) of CSL on Vitamin A consumption during 7 days

Table 4: Climate-smart livestock impact on Vitamin A consumption during 7 days

Vitamin A consumption 7 DAYS	Coef,	Standard error	z
ATET			
Adopt at least one CSL practice (Yes vs No)			
Nearest-neighbor matching	-0.064	0.21	-0.30
Regression adjustment	0.0568	0.1726	0.3300
Inverse-probability weights	0.1060	0.1714	0.6200

Number of observations = 460

During seven days, the result of nearest-neighbor matching, regression adjustment, and inverse-probability weights shows that, regardless of the method. So, the adoption of climate-smart livestock doesn't significantly impact the household's ability to consume a diet that contains vitamin A.

7.9. Treatment-effects estimation (PSM) of CSL on Vitamin A consumption during 24H

Table 5: Climate-smart livestock impact on Vitamin A consumption during 24 hours

Vitamin A consumption 24H	Coefficient	Standard error	z
ATET			
Adopt at least one CSL practice (Yes vs No)			
Nearest-neighbor matching	0.022	0.235	0.09
Regression adjustment	-0.111	0.191	-0.580
Inverse-probability weights	-0.103	0.196	-0.520

Number of observations = 460

As the preview result on the impact of climate-smart livestock adoption on vitamin A consumption during the seven days, the adoption of CSL practices doesn't also influence the score of vitamin A consumption during the last 24 hours. That means the adoption of climate-smart livestock practices doesn't have a significant influence on the household's ability to consume a diet with vitamin A during 24 hours.

7.10. Quality of matching

The figure presents the distribution of propensity scores on the common scale depending on whether the farmer is an adopter or not of climate-smart livestock practices. There is almost identical support for both groups matched. So, we obtained a good match for this propensity. We then conclude that our assumption of common support is validated by the substantial overlap regarding the characteristics of the two groups (treatment and control).

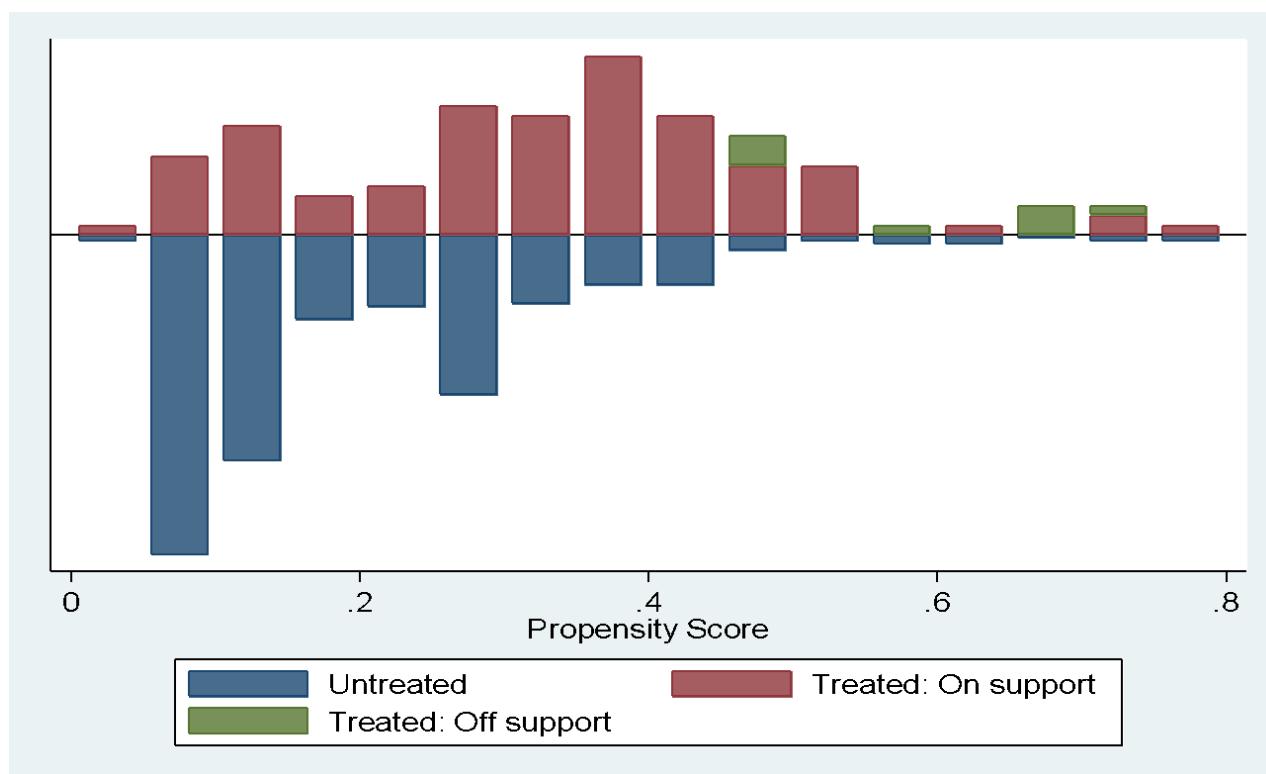
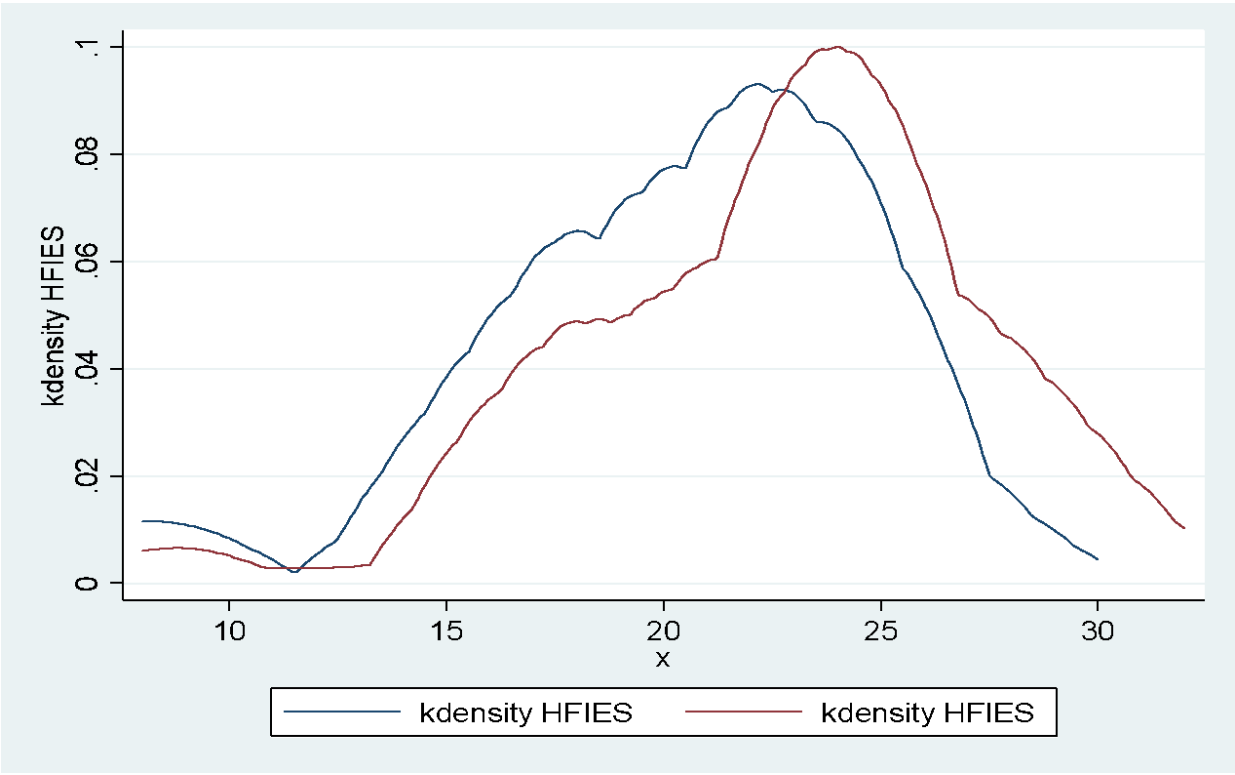
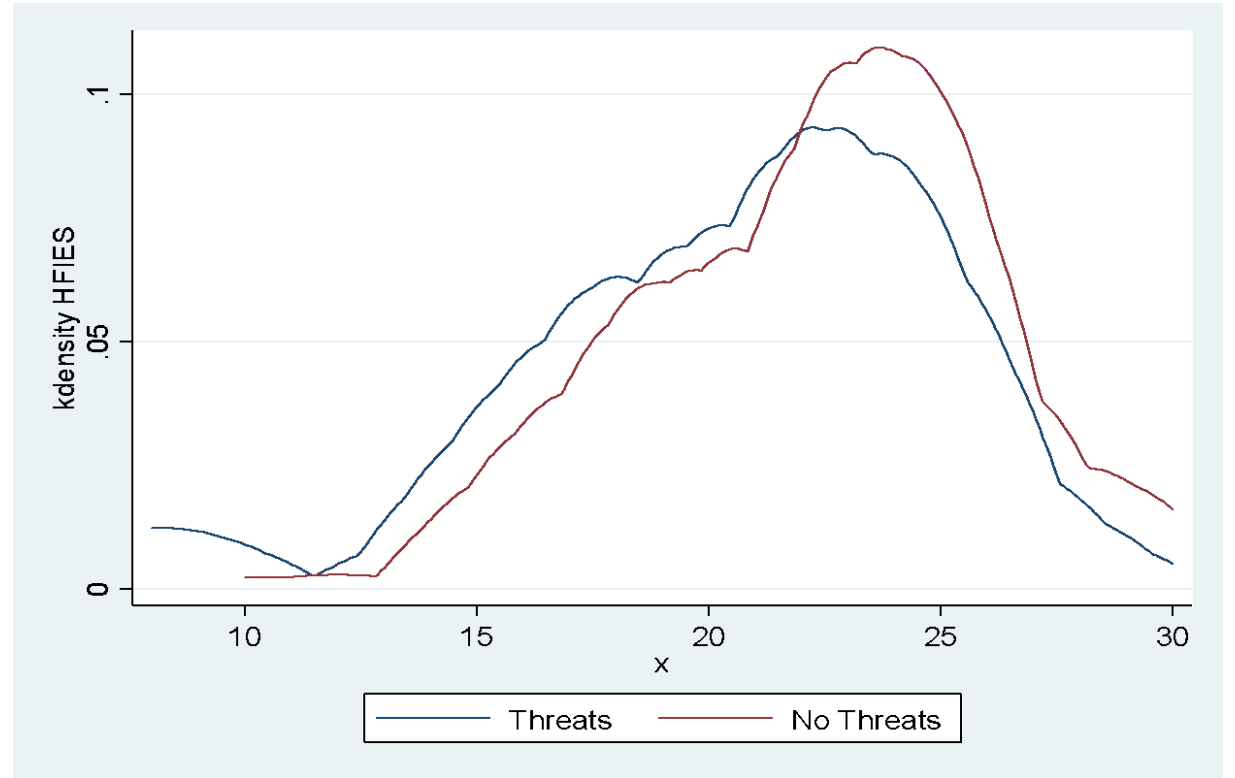


Figure 2: Quality of matching

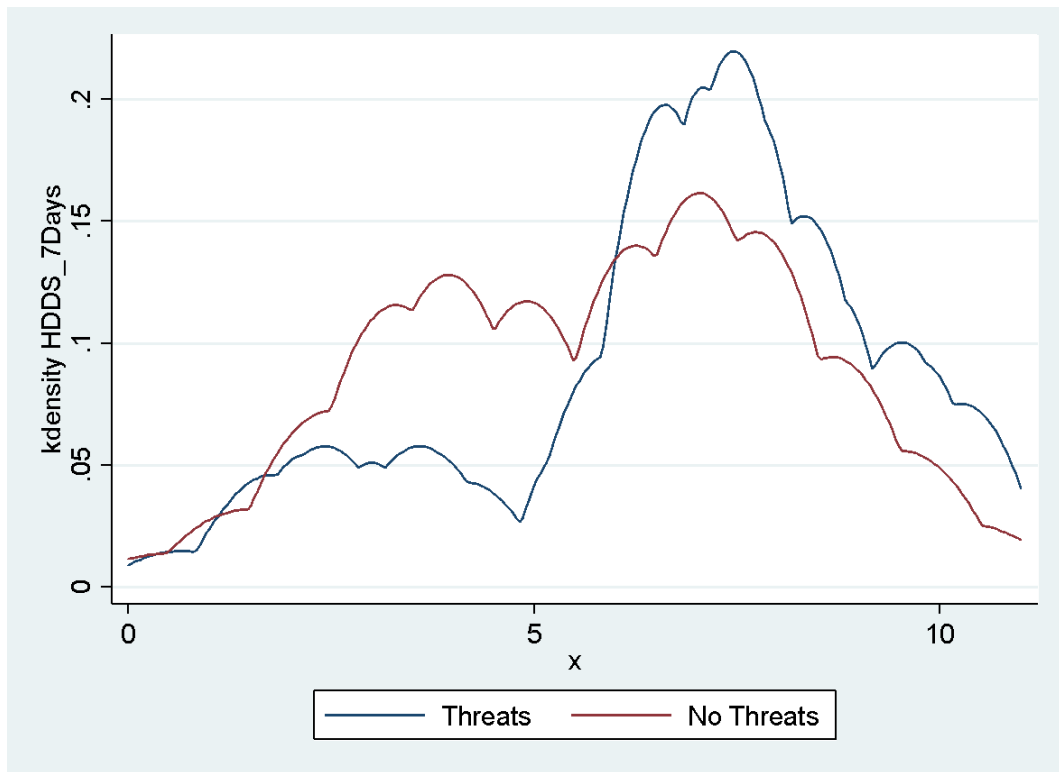
The figures show the kernel density distribution of propensity score matching for the climate-smart livestock adopters and non-adopters before and after the matching. The figure indicates that the propensity score distribution between the two groups significantly improved, and there is an overlap on some points after matching.



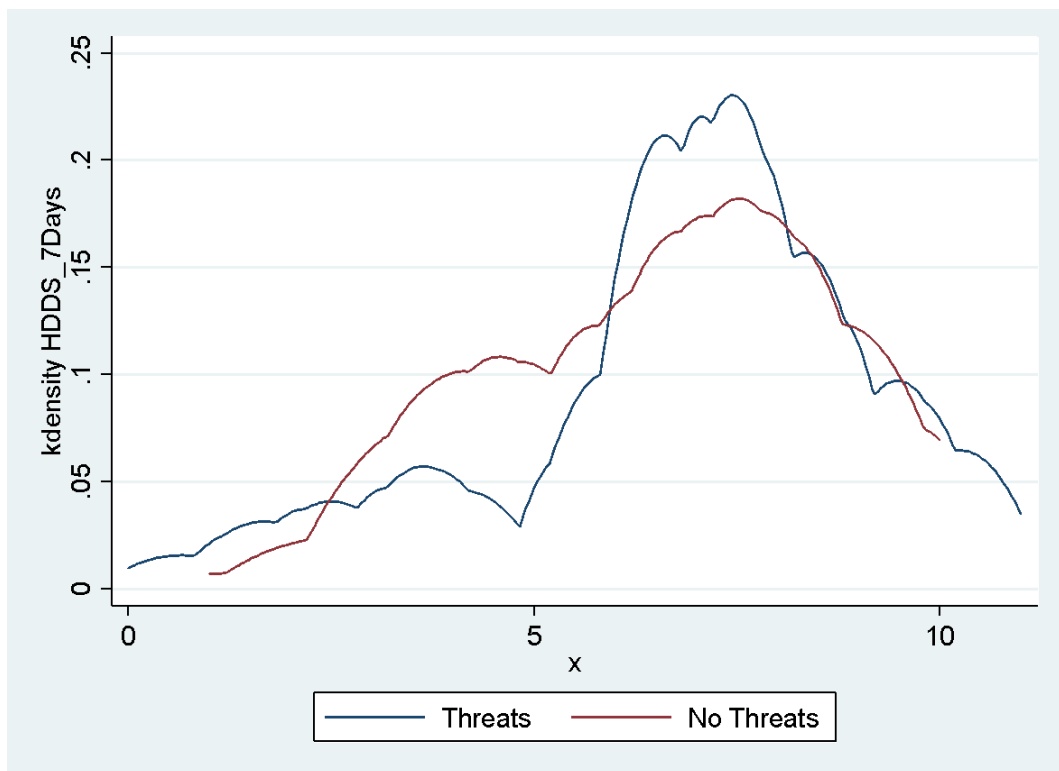
Before



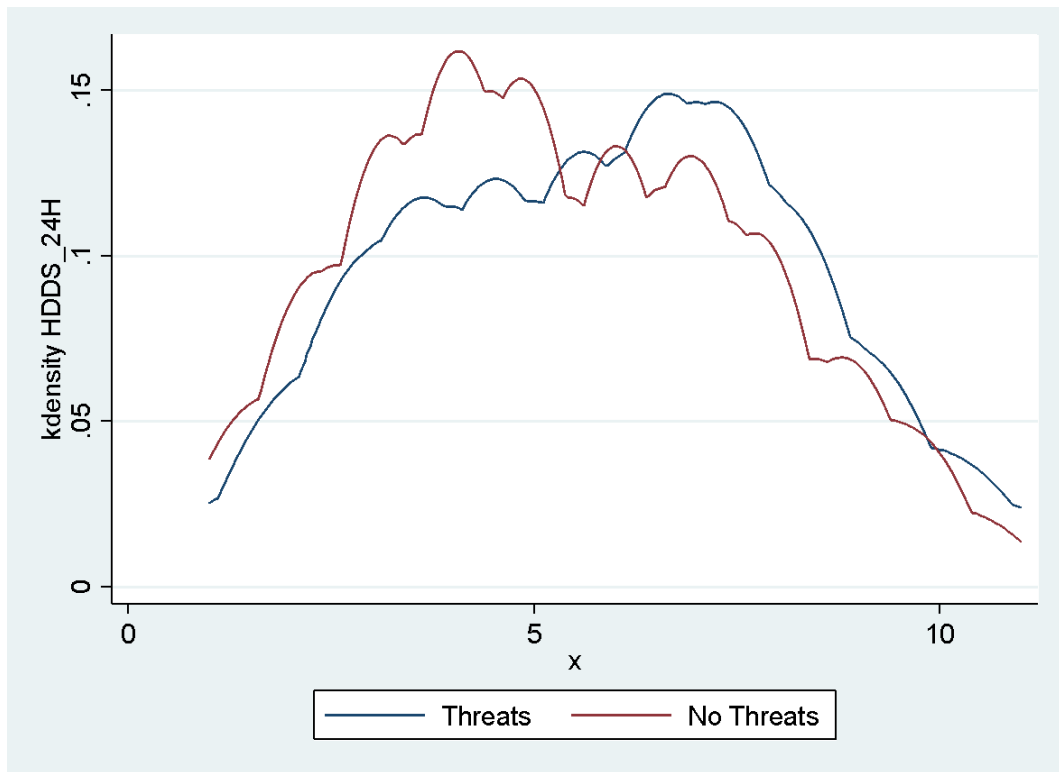
After



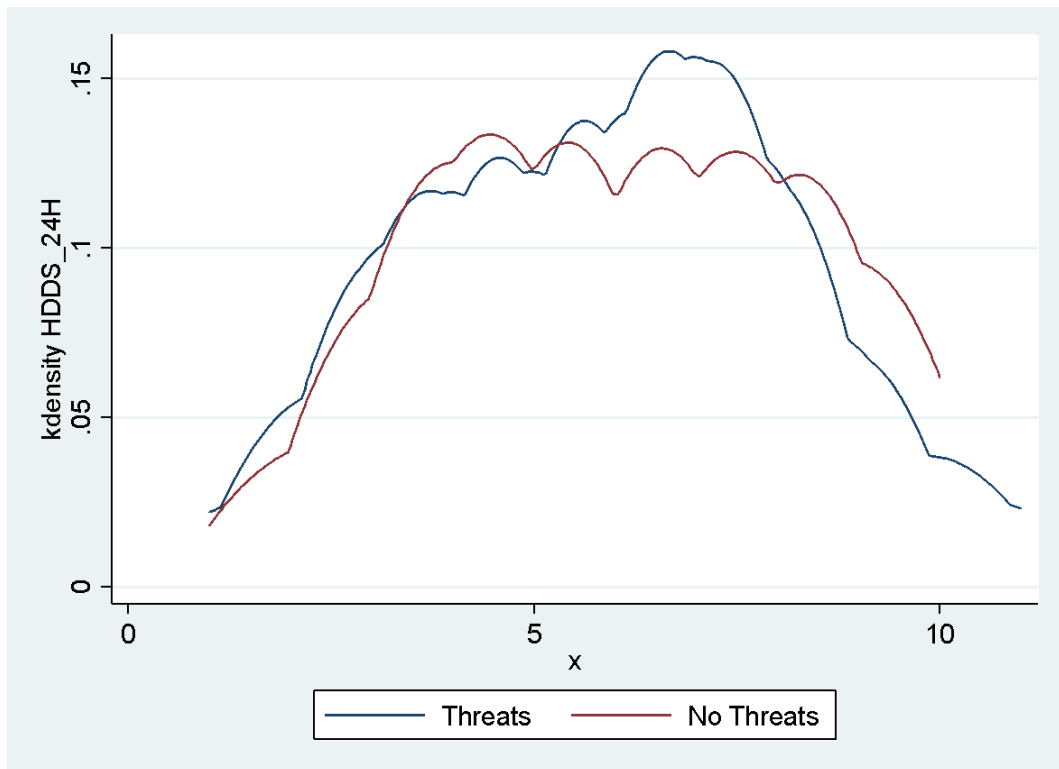
Before



After



Before



After

7.11. Quality measurements of the matching

To determine whether all explanatory variables introduced in the model were adequately balanced and to identify the best-matching estimator, the propensity score test was applied to each matching estimator. Therefore, the joint significance test and the mean bias result confirmed the best propensity matching.

The propensity score and explanatory variable balance test matching estimator in table 6 showed that, before matching, the means of the variables household size, sex, experience in livestock, herd size, access to credit, extension, belonging to a cooperative, and education are significantly different. Nevertheless, after matching, the mean difference between the treatment and control groups is not statistically significant.

Table 6: Performance of the propensity score matching: quality measurements

	Sample	Mean Values		Standardized bias		t-test	
Variable		Treated	Control	% bias	% Bias reduction	t	p> t
Household size	Unmatched	6,282	6,802	-14,200		-1,350	0,178
	Matched	6,363	5,628	20,100	-41,3	1,540	0,126
Sex	Unmatched	0,927	0,920	2,800		0,250	0,799
	Matched	0,922	0,961	-14,700	-422,7	-1,190	0,236
Experience in livestock	Unmatched	8,173	7,739	5,600		0,550	0,583
	Matched	8,147	7,128	13,100	-135,2	0,940	0,349
Herd size	Unmatched	9,627	9,103	3,000		0,310	0,756
	Matched	9,794	8,824	5,500	-85,2	0,380	0,703
Access to credit	Unmatched	0,145	0,043	35,500		3,780	0
	Matched	0,118	0,039	27,200	23,5	2,100	0,037
Extension	Unmatched	0,355	0,140	51,000		5,1	0
	Matched	0,304	0,324	-4,700	90,8	-0,300	0,764
Belonging to a cooperative	Unmatched	0,764	0,418	74,800		6,600	0
	Matched	0,745	0,755	-2,100	97,2	-0,160	0,872
Education	Unmatched	6,400	5,418	24,400		2,300	0,022
	Matched	6,373	6,294	2,000	92,0	0,130	0,895

Sample	PsR2	LR	chi2p>chi2	Mean Bias	MedBias	B	R	%Var
Unmatched	0.126	63.60	0.000	26.4	19.3	92.7*	1.05	50
Matched	0.035	9.82	0.278	11.2	9.3	42.6*	4.00*	75

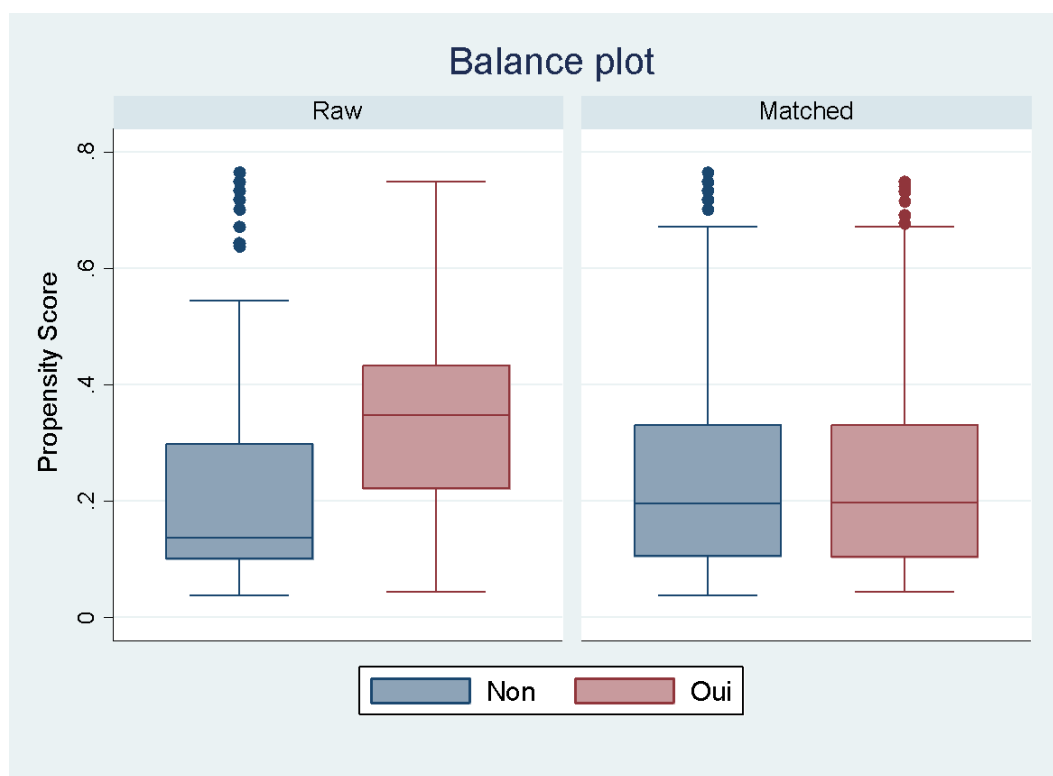


Figure 3: Quality of matching using a boxplot

Figure 3 shows the quality of matching using a boxplot. It's seen a significant difference between threats and no threats in the propensity score before matching. Nevertheless, after matching, the propensity score is the same as it is shown in the figure. So, I obtained a good match using the variables.

7.12. Sensitivity Analysis

Sensitivity analysis assesses the sensitivity of a small variation of the treatment effect estimated in the model's specification, a strong assumption that must be maintained to isolate the pure effect of interventions on outcome variables. In the literature, the recommendation is the bounding methods to verify whether estimates of mean treatment effects on those treated (ATT) are responsive to external changes other than treatment (Rosenbaum & Rubin, 1983). Accordingly, the sensitivity analysis presented in Table below confirms the absence of hidden biases and confirms that the estimated treatment effects for livestock farmers are solely due to the introduction of climate-smart livestock practices. This result is similar to (Fentie & Beyene, 2019; Mossie, 2022)

Table 7: Sensitivity analysis using Rosenbaum bounds for food security

Rosenbaum bounds for HFIES (N =459 matched pairs)

Gamma	Sig + upper bound significance level	Sig-lower bound significance level	t-hat + upper bound Hodges-Lehmann point estimate	t-hat-lower bound estimate Hodges-Lehmann point estimate	CI + upper bound confidence interval (a = .95)	CI-lower bound confidence interval (a = .95)
1	0	0	22.5	22.5	22	23
1.1	0	0	22	22.5	21.5	23
1.2	0	0	22	22.5	21.5	23
1.3	0	0	22	23	21.5	23.5

Gamma	Sig + upper bound significance level	Sig-lower bound significance level	t-hat + upper bound Hodges-Lehmann point estimate	t-hat-lower bound estimate Hodges-Lehmann point estimate	CI + upper bound confidence interval (a = .95)	CI-lower bound confidence interval (a = .95)
1.4	0	0	21.5	23	21	23.5
1.5	0	0	21.5	23	21	23.5
1.6	0	0	21.5	23.5	21	23.5
1.7	0	0	21	23.5	20.5	24
1.8	0	0	21	23.5	20.5	24
1.9	0	0	21	23.5	20.5	24
2	0	0	21	23.5	20.5	24

8. Conclusion

The goal of this study was to analyze the factors influencing the adoption and the intensity of adoption of climate-smart livestock practices among farmers in Benin and the impact of this adoption on food security. An investigation was conducted in North Benin, where livestock is most important, and the effects of climate change are intense. The findings suggest that factors such as experience in livestock, the cattle herd size, and belonging to a cooperative influence positively and significantly the CSL probability of adoption practices at 5 %, respectively, of 0.5 %, 0.2 %, and 8.6%. Also, education (SDG4) and agriculture advices are factors influencing the CSL probability adoption positively and significantly at 1 %, respectively, of 1.1 % and 27.2%. For the intensity of CSL adoption, factors like the contact with extension services, agriculture advices, and agriculture as the first activity (SDG2) influence this intensity.

For the second objective, food security is apprehended in three ways: the Household Food Insecurity Access Score, the Household Dietary Diversity Score, and vitamin A consumption. The last two measures are done in seven days and twenty-four hours. The results show that the adoption of climate-smart livestock practices reduces food insecurity. Therefore, the adoption of one of these practices decreases the household food insecurity access score to 2.131, -1.483, and -1.821, respectively, for the Nearest-neighbor matching, Regression adjustment, and the Inverse-probability weights methods. But, at the same time, the adoption of climate-smart practices doesn't improve the Household Dietary Diversity Score and vitamin A consumption during seven days and 24 hours, with the same methods. The future research can be oriented on the impact of the number practices adopted on the Household Food Insecurity Access Score, the Household Dietary Diversity Score, and vitamin A consumption.

9. Declarations of interest

The authors declare no competing interests.

10. Acknowledgements

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11. Ethical approval

In this research, the dignity, rights, and welfare of respondents are protected.

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